Supervyse IKVM Java Viewer User Guide

Revision 1.14

09/2025

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Revision History

Revision Number	Date	Author	Reviewer	Summary
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1.1	03/07/2019	Jamey Hou	Chijung Lin	Add login/chatroom
1.2	07/28/2020	Yucheng Chen	Chijung Lin	Add S5X features
1.3	08/11/2020	Yucheng Chen	Chijung Lin	Update UI, add issue - Unable to run .jar with double click on RHEL 8.x
1.4	10/08/2020	Yucheng Chen	Chijung Lin	Update UI
1.5	10/23/2020	Yucheng Chen	Chijung Lin	Refine document and fix typo error
1.6	02/22/2021	Yucheng Chen	Chijung Lin	Add maximum length limitation information to ipmi/chatroom window section.
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1.14	09/01/2025	Jamey Hou	Ethan Tseng	Add a description to explain the "Load Default" button.

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1 Introduction

The Java KVM Viewer is a keyboard-video-mouse (KVM) desktop-sharing system that uses the Remote Frame Buffer protocol (RFB) to remotely control another computer. It transmits the keyboard and mouse events from one computer to another, relaying the host screen updates back in the other direction, over a network.

1.1 Overview

The document helps the user to pick up what Java KVM viewer is and how to use it.

1.2 Related Information

Notice: This application is implemented in Java**,** so JRE 8+ is required.

1.3 Terms

The following terms are used throughout this document:

Java KVM Viewer

A Java based application to provide Insyde KVM interface to access host.

Host

The target machine that user wants to operate on.

ikvmserver

The server-side daemon on BMC. It's used as an entry to let user remote desktop the host.

Video frame

A basic frame unit to get from the host screenshot.

Keyboard event

The keyboard events are triggered when users key in. Each keyboard event is used to operate the host's keyboard.

Mouse event

The mouse events are triggered when users move the mouse. Each mouse event is used to operate the host's cursor.

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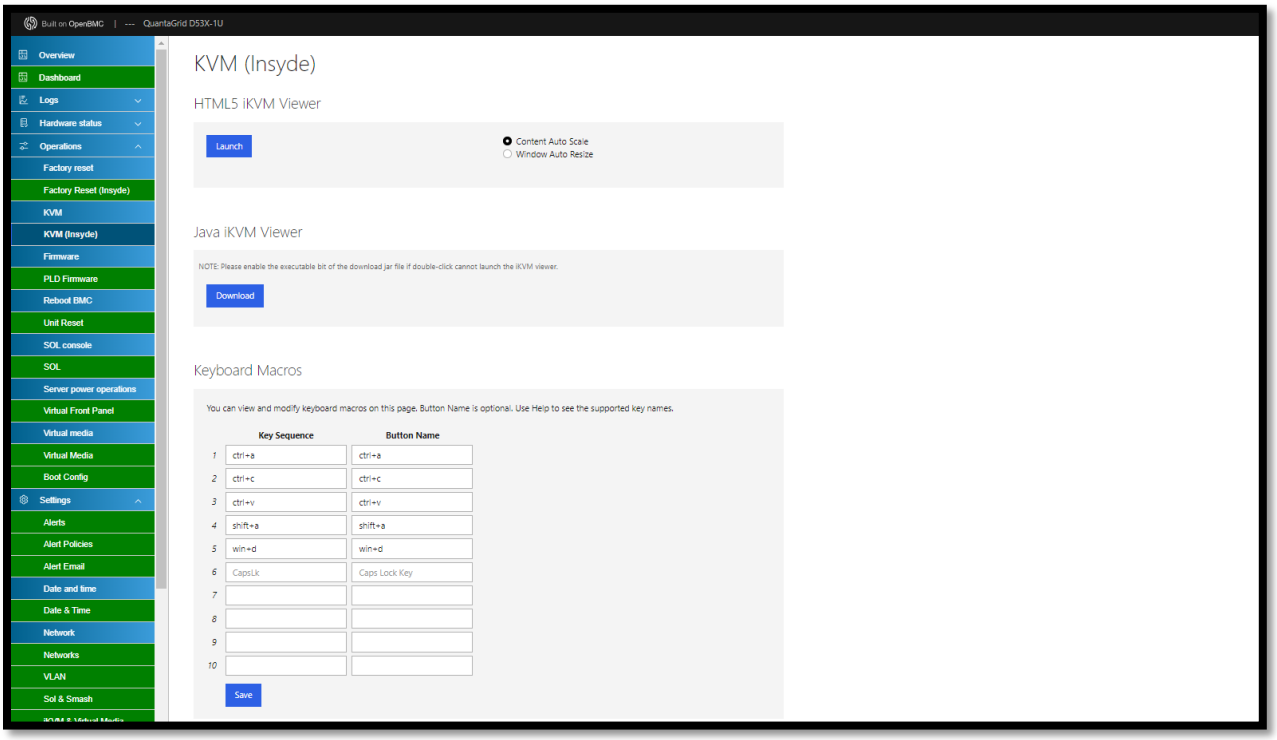
2 Quick to Start

2.1 Download the iKVMViewer.jar

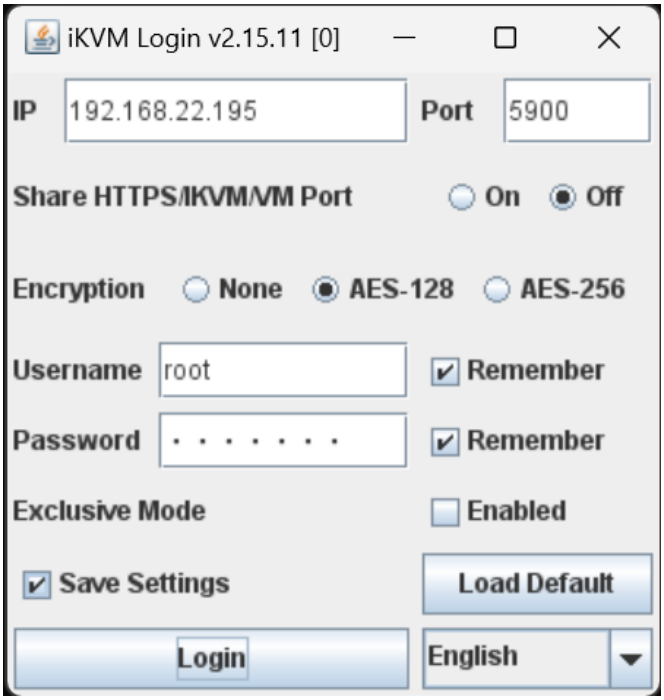
1. Login BMC web.

2. Go to page Operation > KVM.

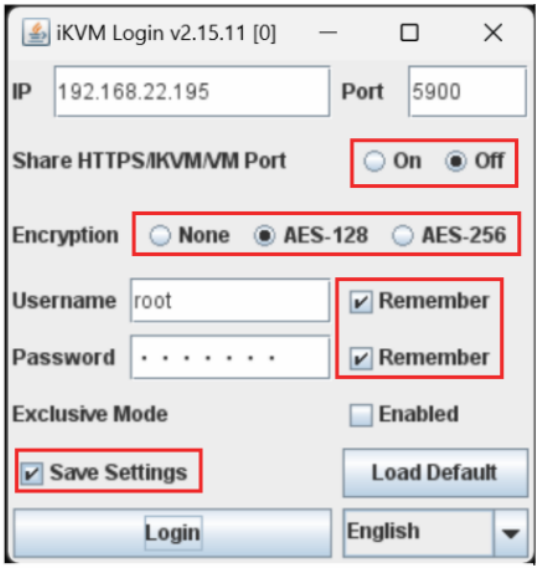


3. Click the **"Launch"** button under **"Java iKVM Viewer"** and download.

4. Double click to launch the **iKVMViewer.jar**.

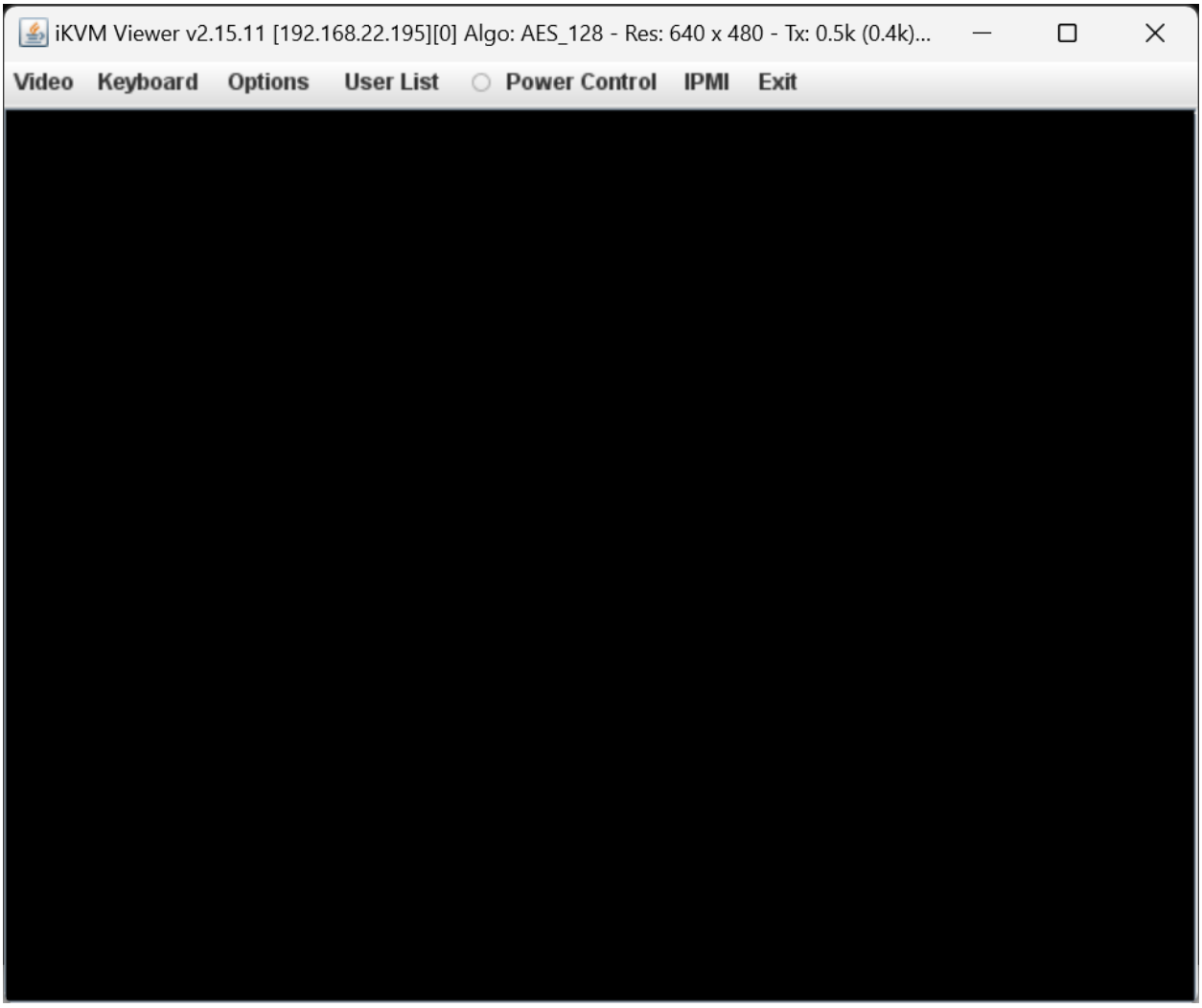


- You will see a login screen like this. Fill in the required **IP**, **Port**, **Username**, and **Password** fields.
- If **Share HTTPS/IKVM/VM Port** was configured to **On**, please fill in the port HTTPS (Web) using to **Port**.
- If **Exclusive Mode** is **Enabled**, only one user can use KVM.
- When clicking **Load Default**, the fields outlined in red will be reset.

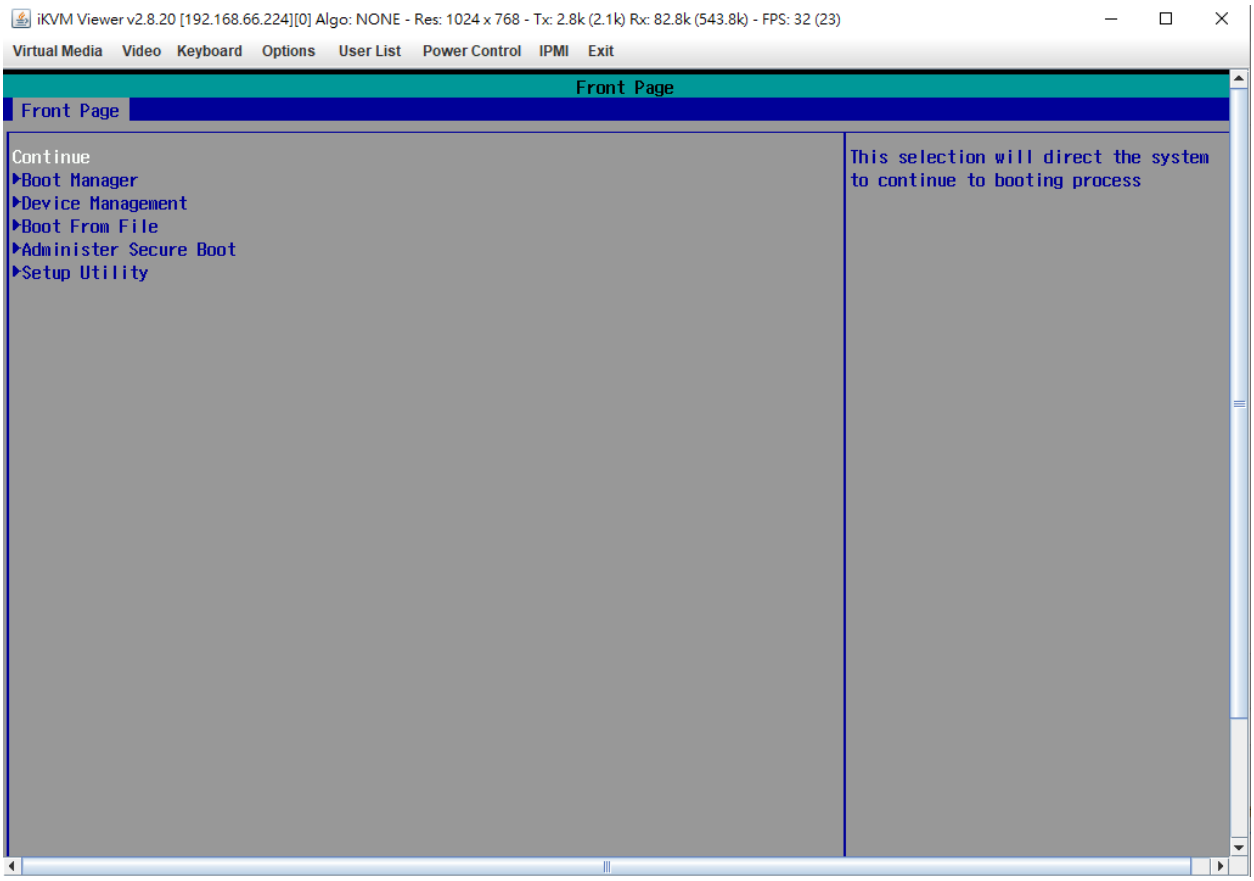


5. Press the "Login" button to connect to KVM server.

6. After login, the main window of KVM viewer will display.



2.2 Basic Information



- Title information
  - **"v2.8.20"** is the iKVMViewer program version.
  - **"[192.168.66.224]"** is the server IP.
  - **"[0]"** is platform ID. (default is 0)
  - **"Algo: NONE"** means encryption type is NONE.
  - **"Res: 1024 x 768"** is the video resolution in 1024x768.
  - **"Tx: 2.6k Rx: 2.8k"** is the network stream bandwidth (unit: bps).
  - **"FPS:32 (23)"** is the video fps and average fps.

2.3 KVM Function

- Keyboard/Mouse Redirection
  - Redirect keyboard event to Host.

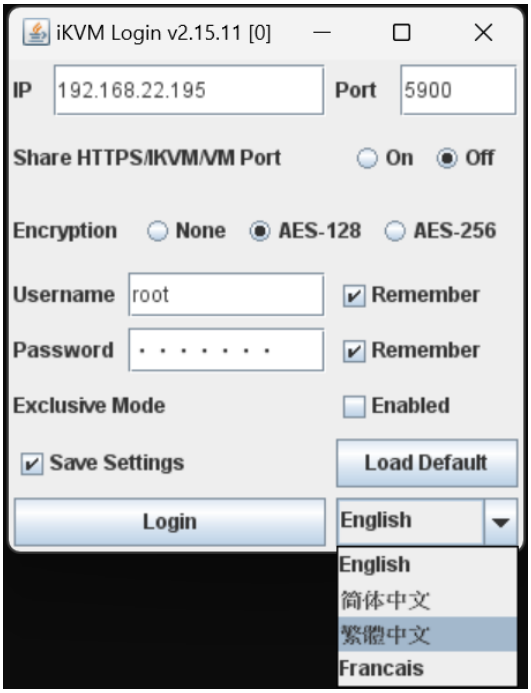
- Redirect mouse event to Host.
- Only the first login owner keeps the keyboard/mouse privilege. When other client wants to get the privilege

- Video Redirection
  - Redirect video frame from Host.
  - Max resolution: 1920x1080.

2.4 Multi-Lang

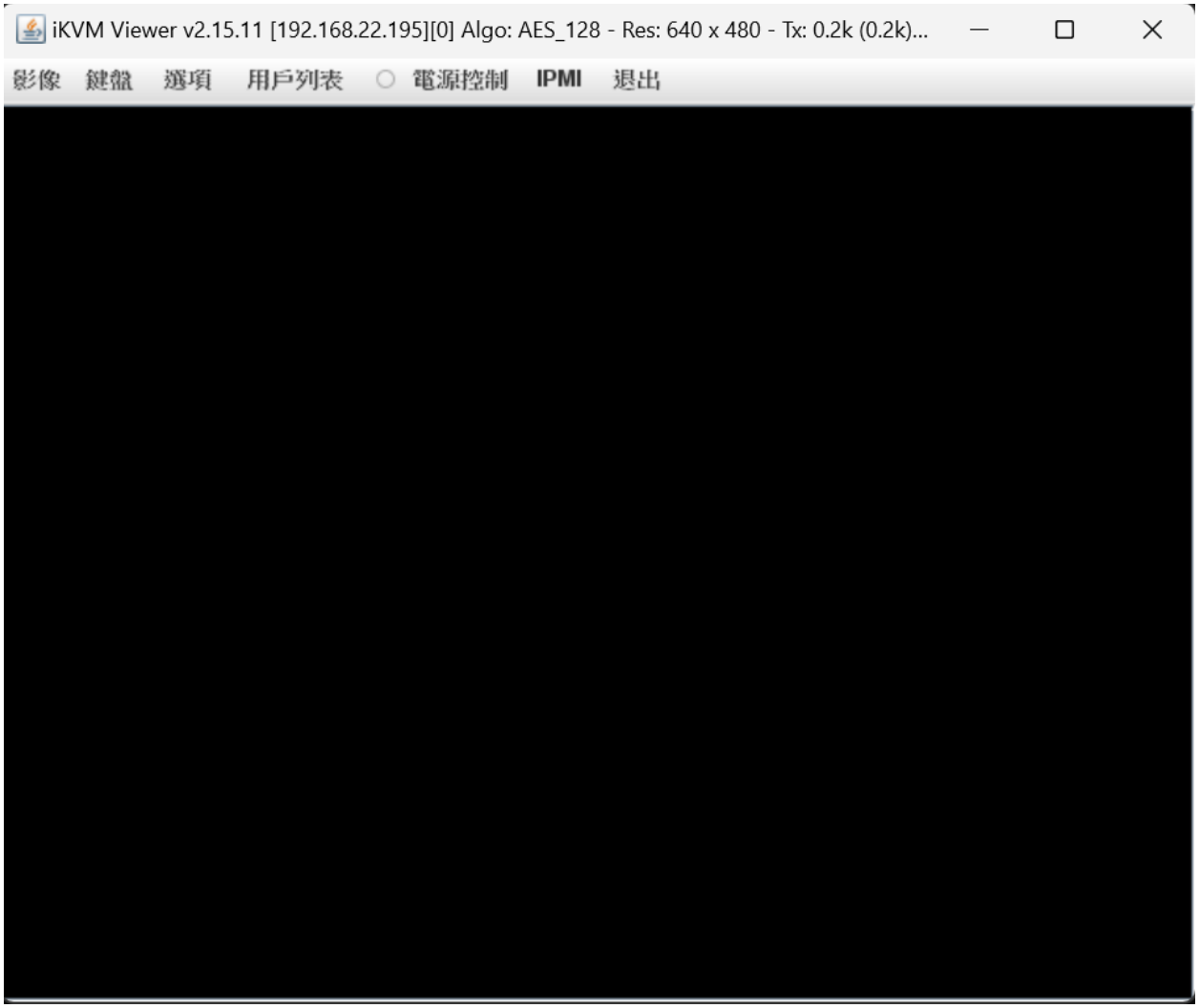
Java Viewer supports several language UI. Launch the Java KVM viewer, there is a selection bar to change the language of UI.

- **English** (default)
- 简体中文
- 繁體中文
- **Francais**



Try to select 繁體中文, and the UI will be changed.

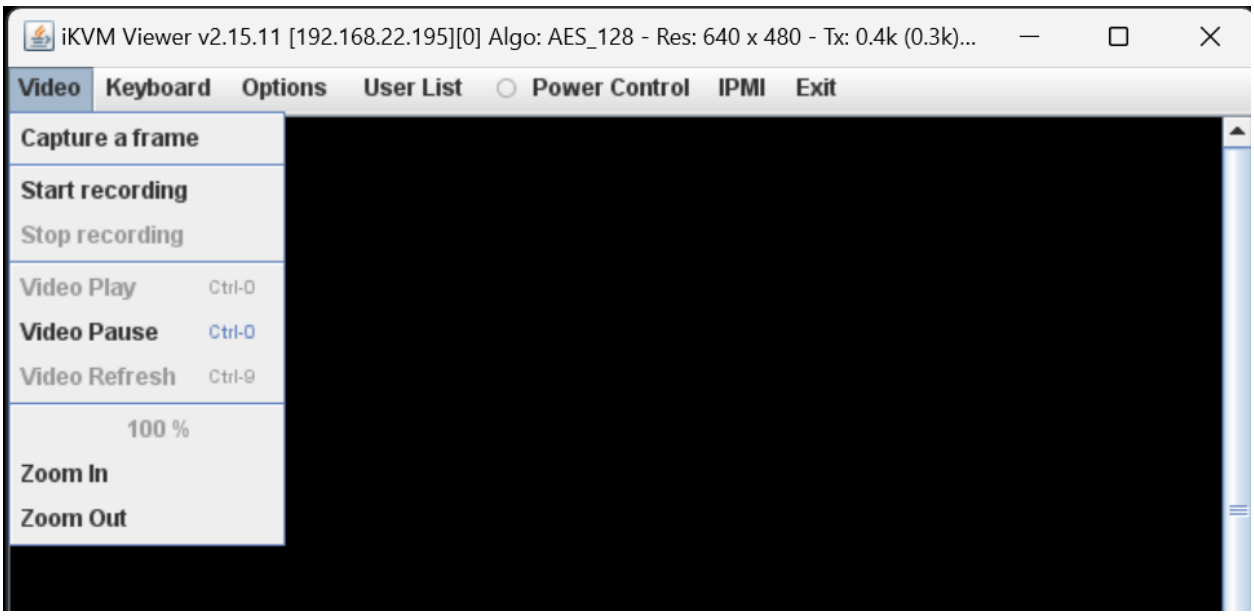
iKVM 登入 v2.15.1...
IP
192.168.22.195
埠
5900
共用 HTTPS/IKVM/VM 埠
開
關
加密
無
AES-128
AES-256
帳號
root
記住
密碼
記住
獨占模式
Enabled
保存設置
載入預設值
登入
繁體中文


iKVM Viewer v2.15.11 [192.168.22.195][0] Algo: AES_128 - Res: 640 x 480 - Tx: 0.2k (0.2k)...
影像
鍵盤
選項
用戶列表
電源控制
IPMI
退出

3 Advance Setting

3.1 Video

Video related features including video flow control, frame capture, and recording.



3.1.1 Flow control

Provide video flow control to play/pause/refresh video frame as media player.

- **Pause** - Freeze the video.
- **Play** - Continue to play video.
- **Refresh** - Get a new video frame. (Only able to be used when the video flow is in pause status.)

3.1.2 Frame Capture

- **Capture a Frame** - Click to take a screenshot and save it as a .jpg format image.

3.1.3 Recording

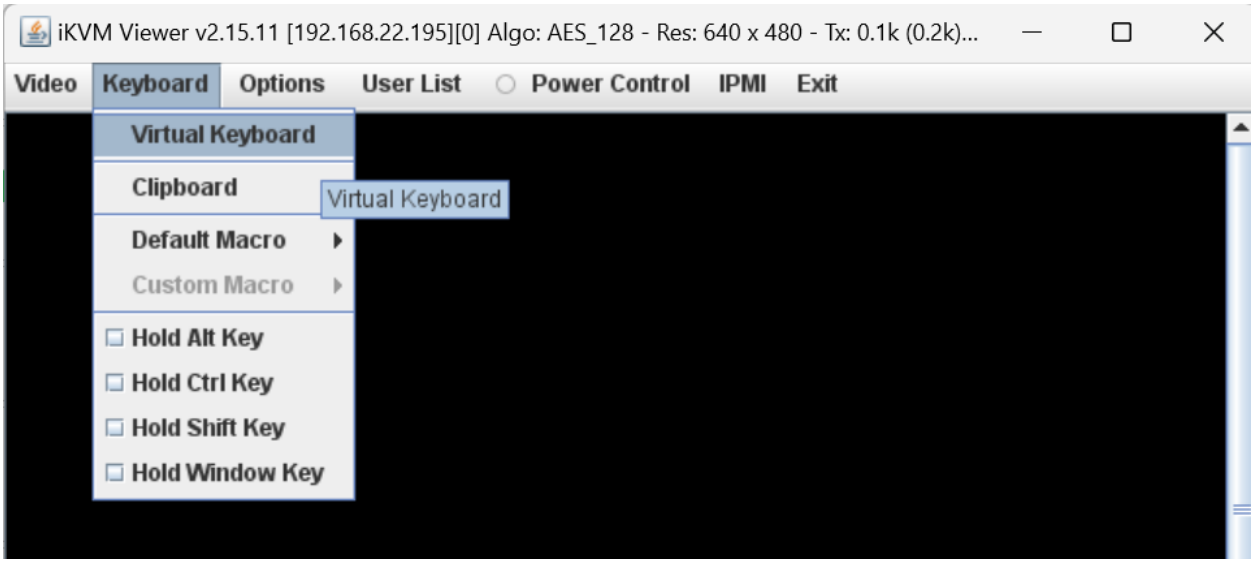
Video recording provides users to record video from the viewer's screen. The recording result will be saved as a .avi file. Click **"Start Recording"** to start and then click "**Stop Recording**" to finish.

- **Start Recording** - Start a video recording.
- **Stop Recording** - Stop recording.

- The length of recording time is 2 minutes by default. Users can enabled/disabled or modify the recording time at Menu > Options > Preference > Display > Recording time. And the time range is 1 minute ~ 180 minutes.

- **Notice**: If user stores video file to network storage, viewer should wait more time for response. (Ex: set 60s for auto stop, and it will delay about 15s to response.)

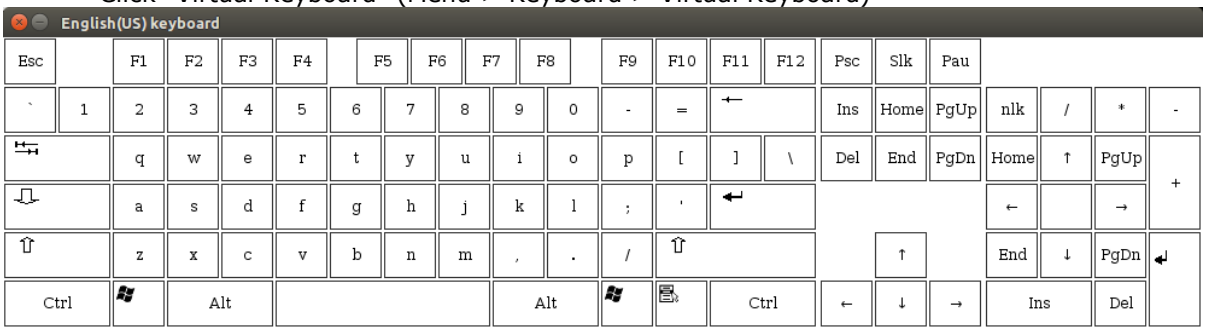
3.2 Keyboard



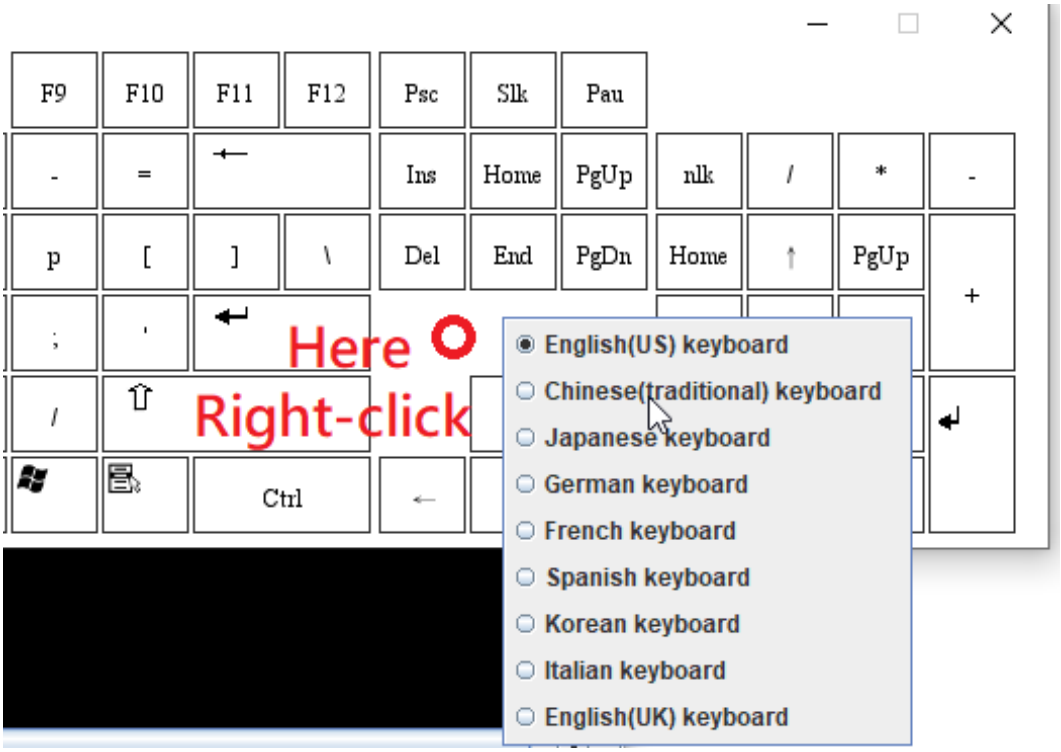
3.2.1 Virtual Keyboard

Provide a UI to let users send key events with a cursor.

- Click "Virtual Keyboard" (Menu > Keyboard > Virtual Keyboard)



You can change the keyboard layout by right-clicking on the keyboard panel blank area to pop-up switch menu.



3.2.2 Keyboard Macro

3.2.2.1 Default Keyboard Macros

There are many available default hot keys.

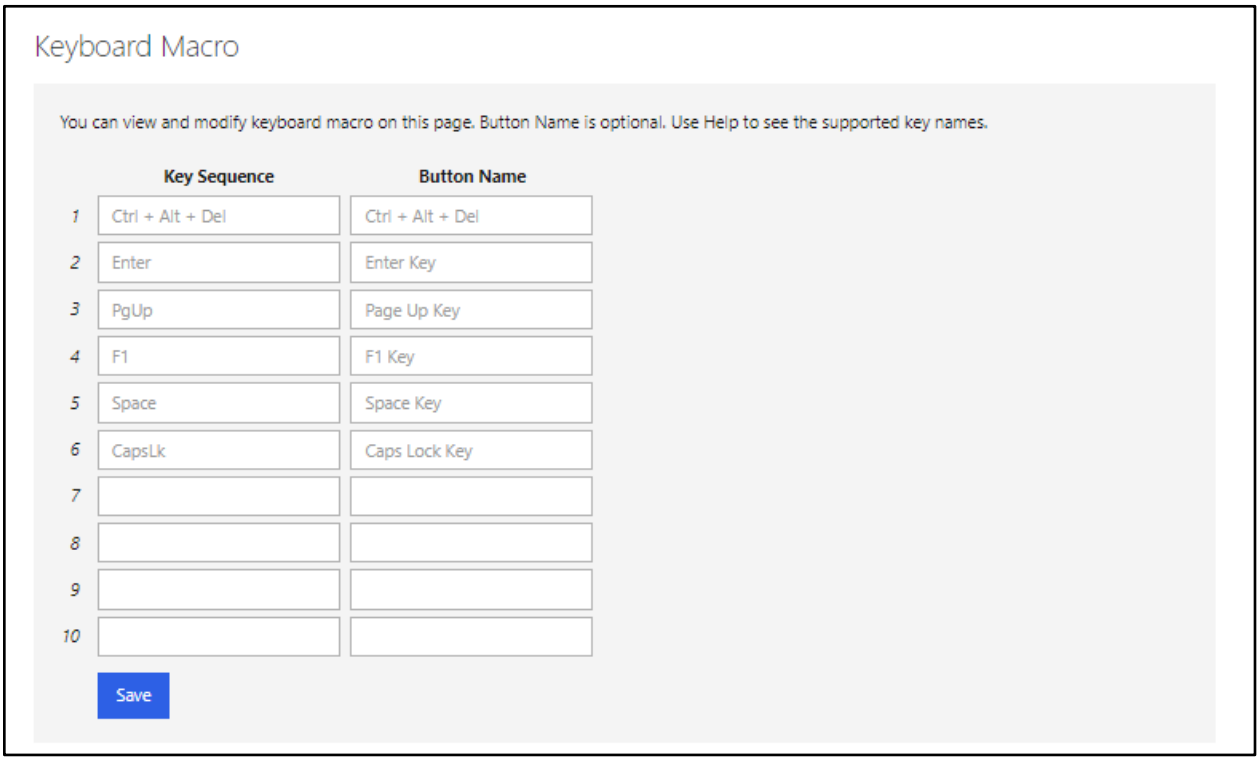
- Default keyboard macros:
    - Ctrl+Alt+Del
    - Alt+Tab
    - Alt+Esc
    - Ctrl+Esc
    - Alt+Space
    - Alt+Enter
    - Alt+Hyphen
    - Alt+F4
    - Alt+PrntScrn
    - PrntScrn
    - F1
    - Alt+F1

- Pause

3.2.2.2 Custom Keyboard Macros

Users can add preferred key macro as custom keyboard macros.

- Key macro management UI: BMC web > Operations > KVM



- **Key Sequence** - Sequence of key events to playback when the macro button is pushed.'+' between keys indicates the keys should be held down together Examples: Ctrl+Alt+Del
- **Button Name** - The key-macro name shown on viewer's custom key-macro fields.
- **Keys** - Key names are the character of keyboard layout such as "a", "5", "\", etc. or else one of the reserved keys in the table below. Names in parentheses are aliases for the same key. Numeric keypad keys are prefixed with "NP_".

Table 3.2.2.2-1 xxxx

Shift (LShift)	RShift	Ctrl (LCtrl)	RCtrl
Alt (LAlt)	RAlt (AltGr)	Win (LWin)	RWin
Enter	Esc	F1 - F12	
Bksp	Tab	CapsLk	Space
Ins	Del	Home	End
PgUp	PgDn	Context (Menu)	
Up	Left	Down	Right
NumLk	NP_Div	NP_Mult	NP_Minus

NP_Plus	NP_0 - NP_9	NP_Dec	NP_Enter
PrtSc (SysRq)	ScrLk	Pause (Break)	

- **Notice**: key sequences are sent to the target as scan codes that get interpreted by the target OS, so they will be affected by modifiers such as CapLock as well as the OS keyboard language setting.

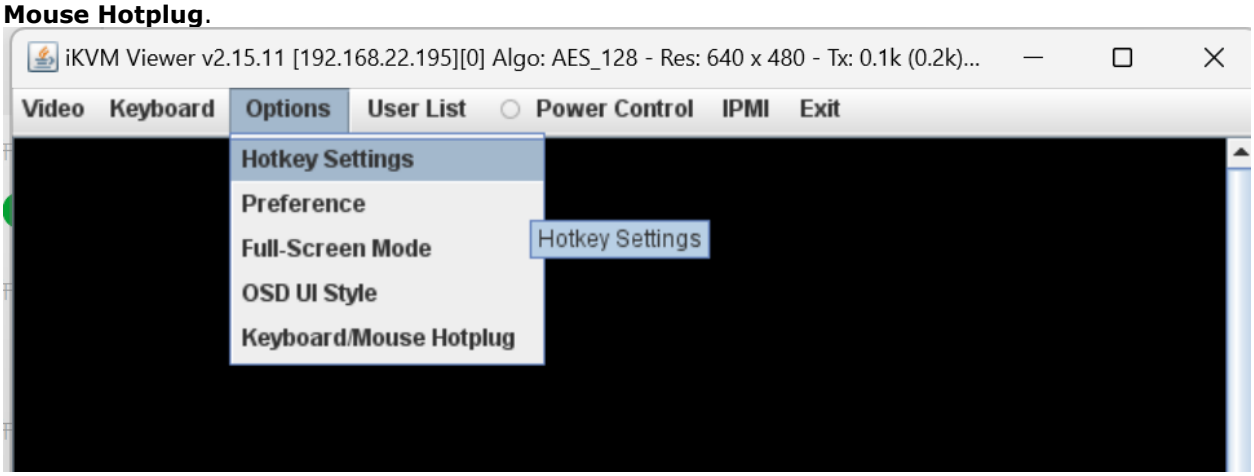
3.2.3 Function key holding

This feature simulates the function key is holding. However, only work with key event rather than mouse event.

- Hold Alt Key
- Hold Ctrl Key
- Hold Shift Key
- Hold Window Key

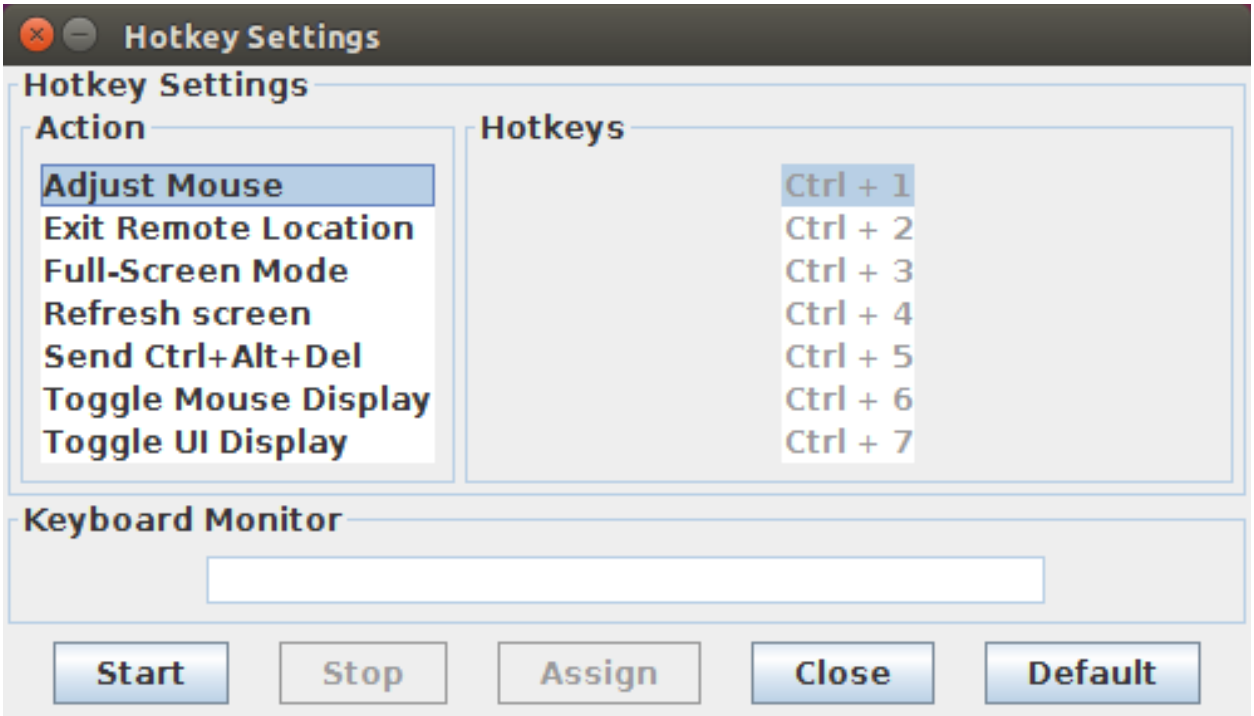
3.3 Options

Features for **Hotkey Settings**, **Preference**, **Full-Screen Mode**, **OSD UI Style**, and **Keyboard Mouse Hotplug**.



3.3.1 Hotkey Settings

- Users can use a hotkey to trigger the Java Viewer feature. Such as toggle local cursor display status.



- Users can modify hotkey by their requirements.

  1) **Start:** Start monitor keyboard input. Users can define their own hotkeys.
  2) **Stop**: Stop monitor keyboard input.

  3) **Assign**: Assign the monitored key to the selected action.

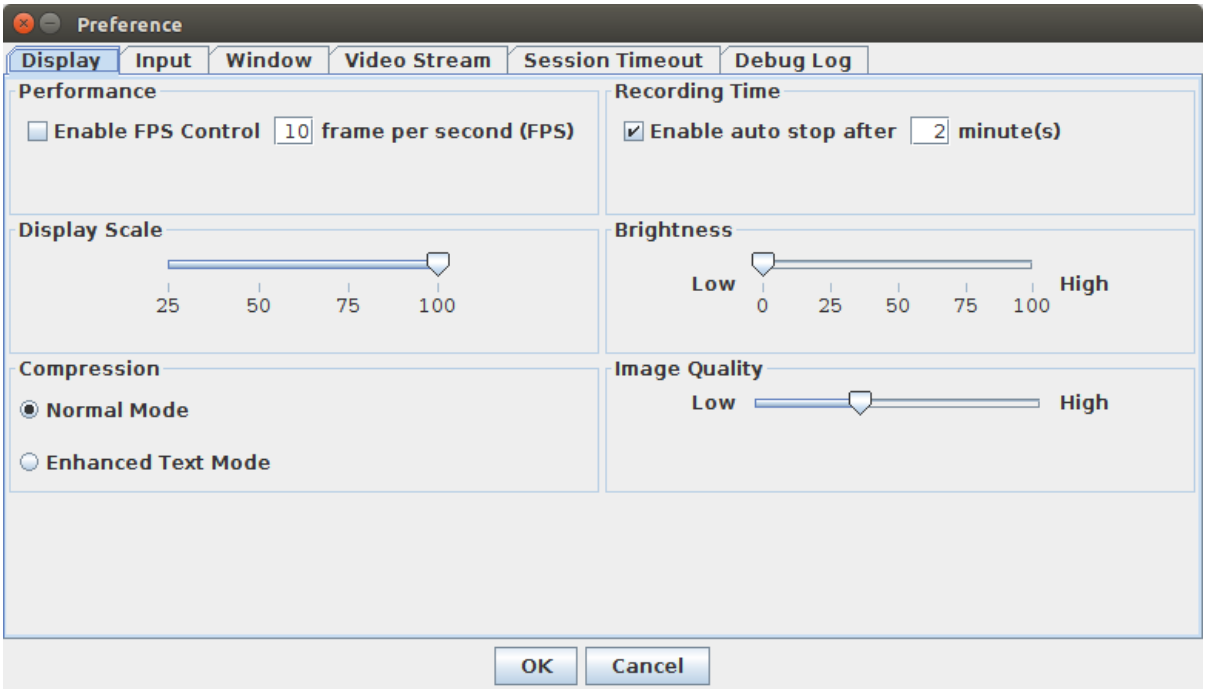
  4) **Close**: Close this panel.

  5) **Default**: Resets all hotkeys to their default values.

- **Notice:** the single function key (ctrl/alt/shift) can NOT be assigned in HotKey Settings. It should be coupled with a normal key.

3.3.2 Preference

3.3.2.1 Display



1) Performance: Enable and set how many frames per second to control display performance.

2) Recording Time: Enable and set recording auto stop.

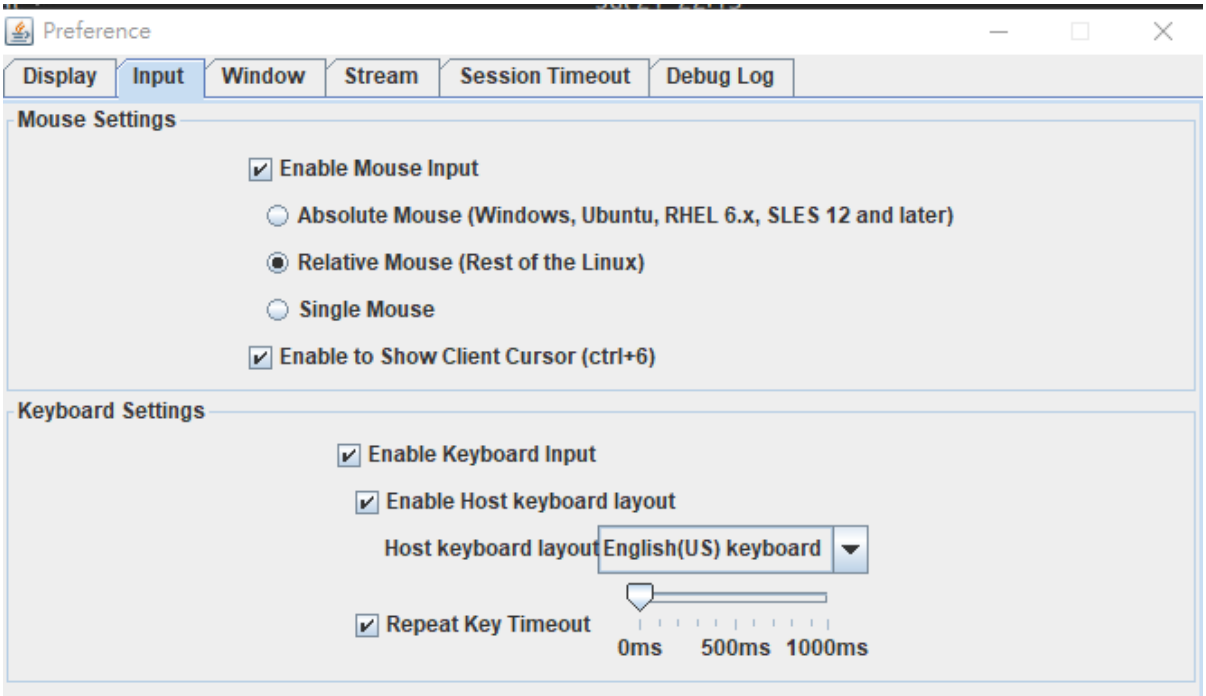
3) Display Scale: Allow to scaling video in 25% ~ 100%.

4) Brightness: Allow to set the brightness of image quality.

5) Compression: Normal Mode (YUV420) / Enhanced Text Mode (YUV444)

6) Image Quantity: Allow to set the brightness of image quality.

3.3.2.2 Input



1) Mouse Settings

- **Enable Mouse Input** - Enable to send client cursor event to Host.
- Able to switch Host cursor mode. Each mode effects the mouse addressing calculation:
  - **Absolute mode** - use the absolute location on screen to address.
  - **Relative mode** - use the bias value to move Host cursor (Ex: X axis +1 means move to leave a unit).
  - **Single mode** - behavior like **Relative mode**. But the client cursor is hidden to avoid user confusion with Host cursor image.
- **Enabled to show client cursor** - show or hide client cursor. The **Single mode** hides the cursor by default.
- **Notice**:
  - Use **absolute mode** first. When **absolute mode** does not work, use candidate **relative/single mode** to move the cursor. (e.g.: bios environment)
  - **Single mode** can be regarded as enhanced **relative mode**. It provides a better user experience.
  - Enter and exit single mode
    - **Enter**: Click the viewer GUI to enter.
    - **Exit**: press combo key **ctrl+6** to leave.

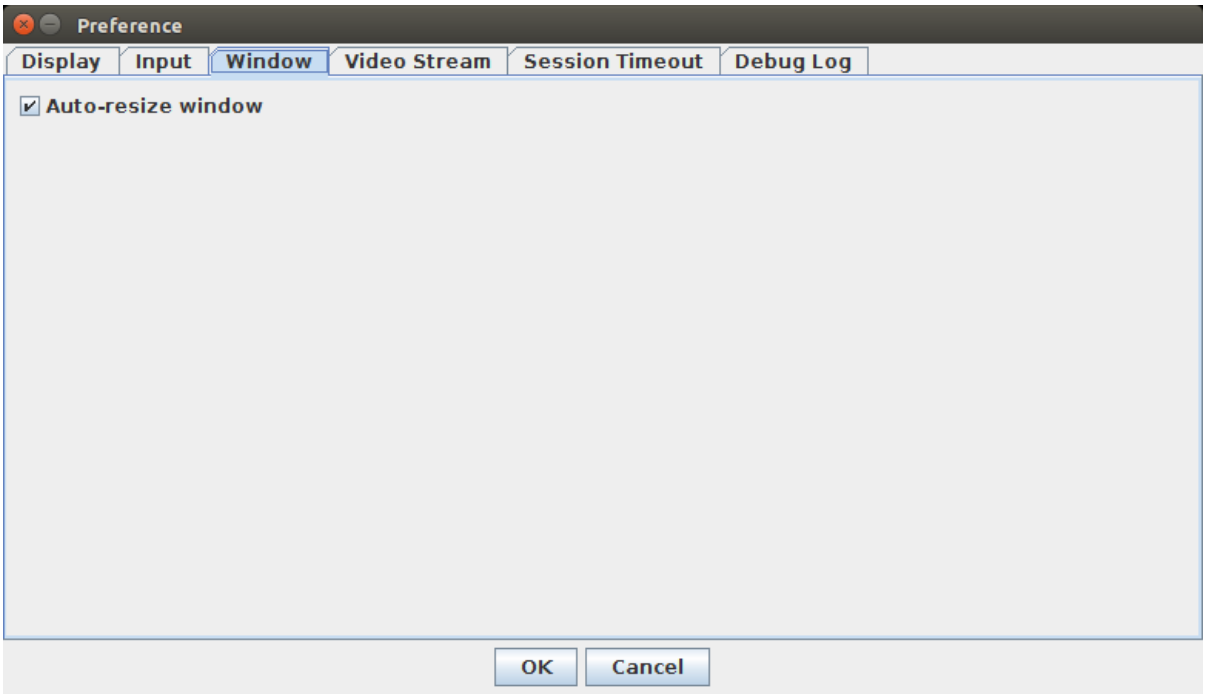
2) Keyboard Settings

- **Enable Keyboard Input** - Enable to send client keyboard event to Host.

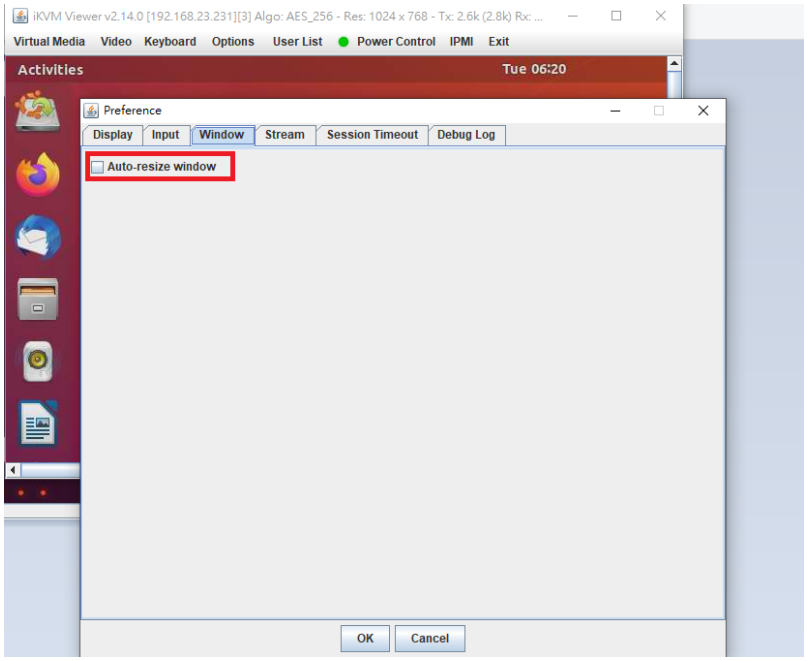
- **Enabled Host Keyboard Layout** - Enabled the host keyboard layout mapping feature. (Please see #4.8 Host Keyboard Layout Mapping Feature)
    - **Host Keyboard Layout** - Adjust the keyboard layout as the same as Host to let ikvmserver select a mapping table.
- **Repeat Key Timeout** - When user presses the keyboard, program detects and sends the event to host per timeout time.

Notice: for a better user experience, the delay time between the 1st and 2nd key event should be at least 0.3s.

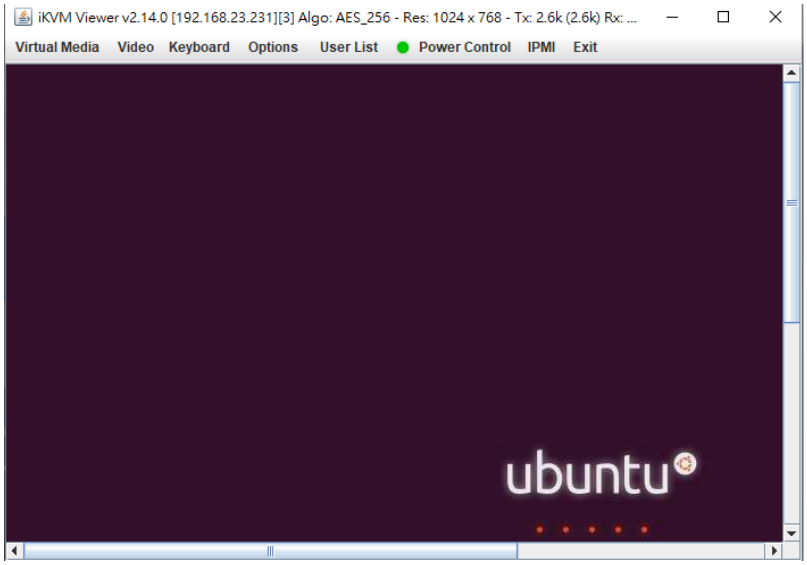
3.3.2.3 Window



- **Auto-resize window** - Enabled/Disable to auto-resize window. When the host resolution is changed, window resize feature will be triggered. In default, window is auto resized.



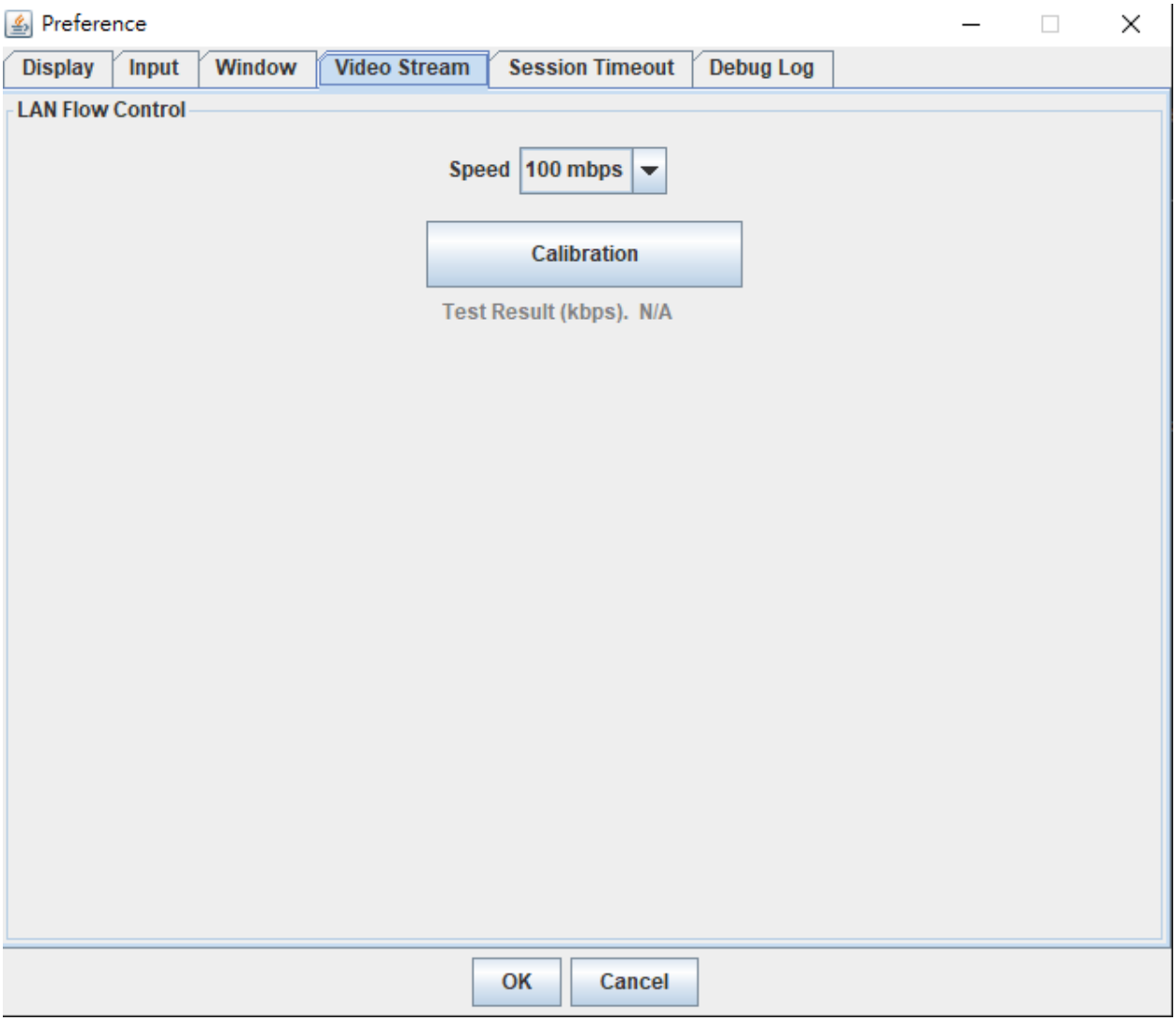
- When the host resolution changes, Java viewer will be the same window size.



3.3.2.4 Video Stream

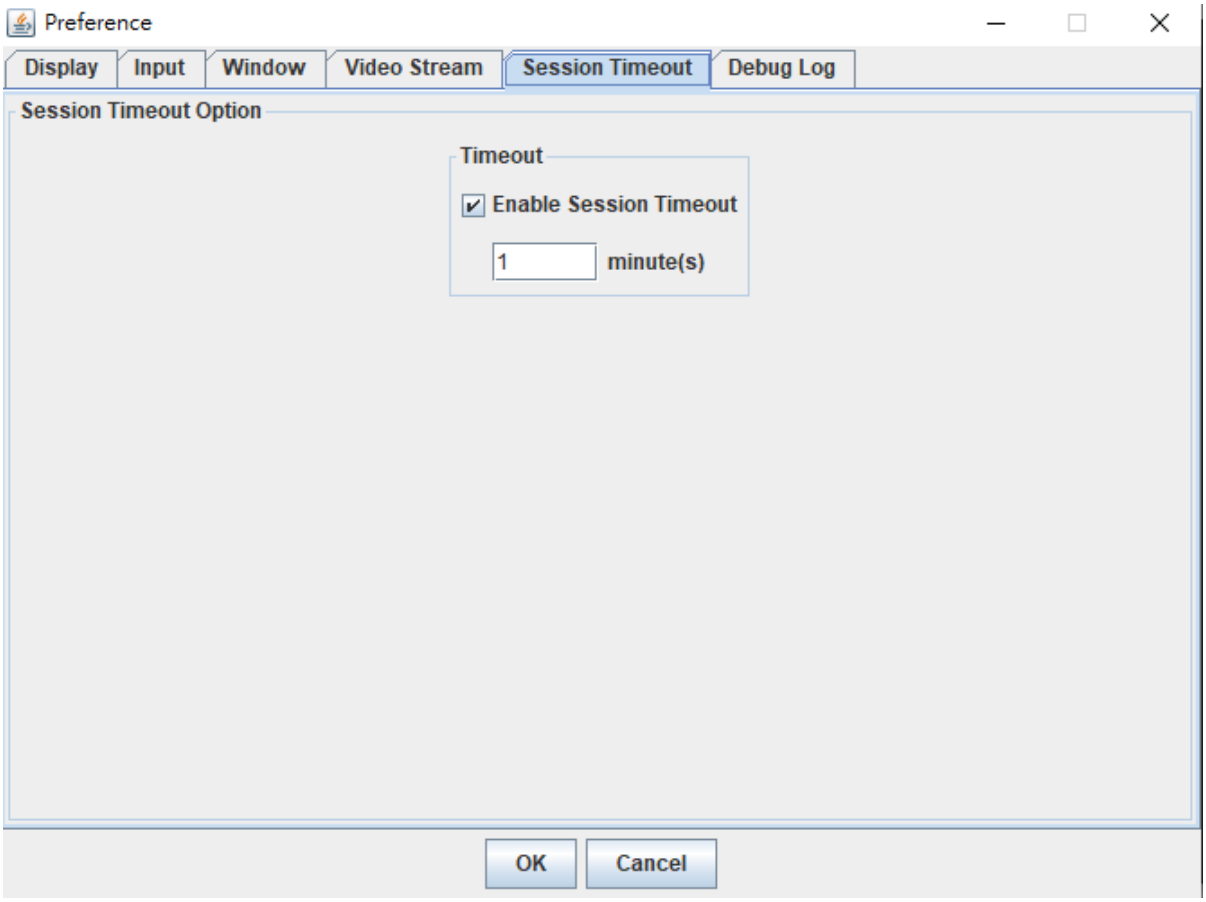
- Setup bandwidth. Available speed: No Limit(default) / 256kbps / 512kbps / 1mbps / 10mbps / 100mbps.
- Support bandwidth calibration.

    1. Press the Calibration button to start.
    2. Wait 10 seconds for the test.
    3. And then the result is shown below.



3.3.2.5 Session Timeout

- When user idles (no keyboard/mouse action) for a timeout time, the session will be expired, and the viewer will disconnect from the server.

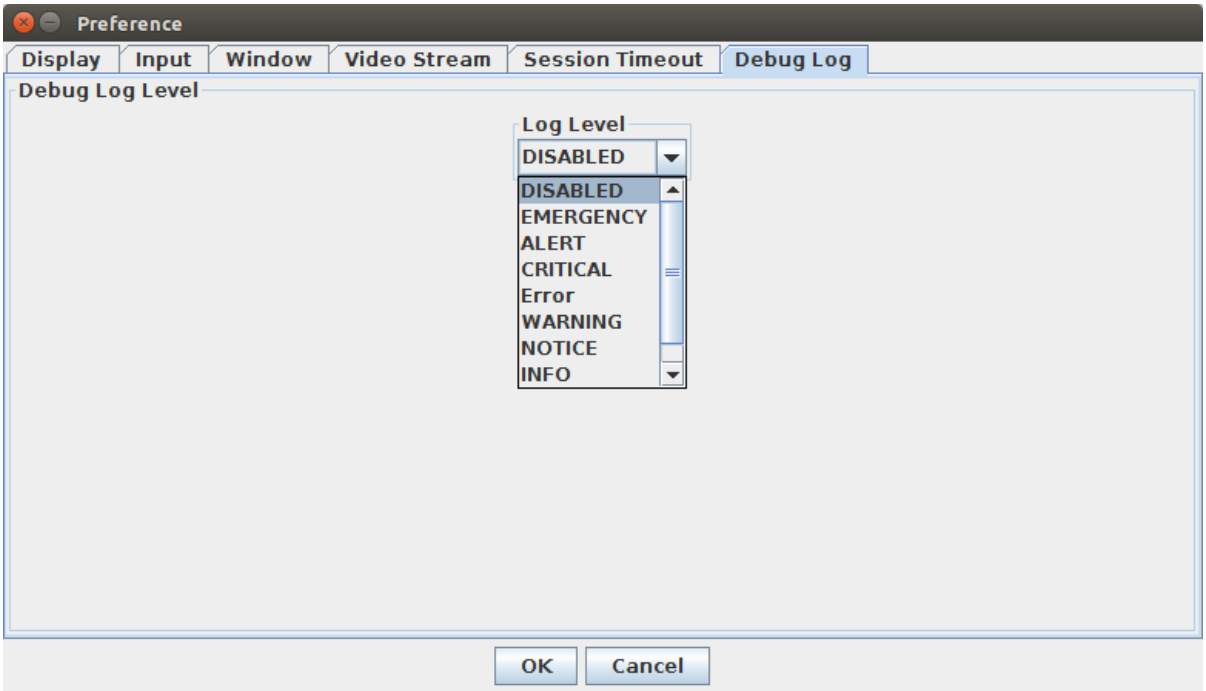


- **Enable session timeout** - Enable and setting session timeout.
    - When session timeout = 0 or disabled, it means no expired time.
    - The unit of session timeout time is in minutes.

3.3.2.6 Debug Log

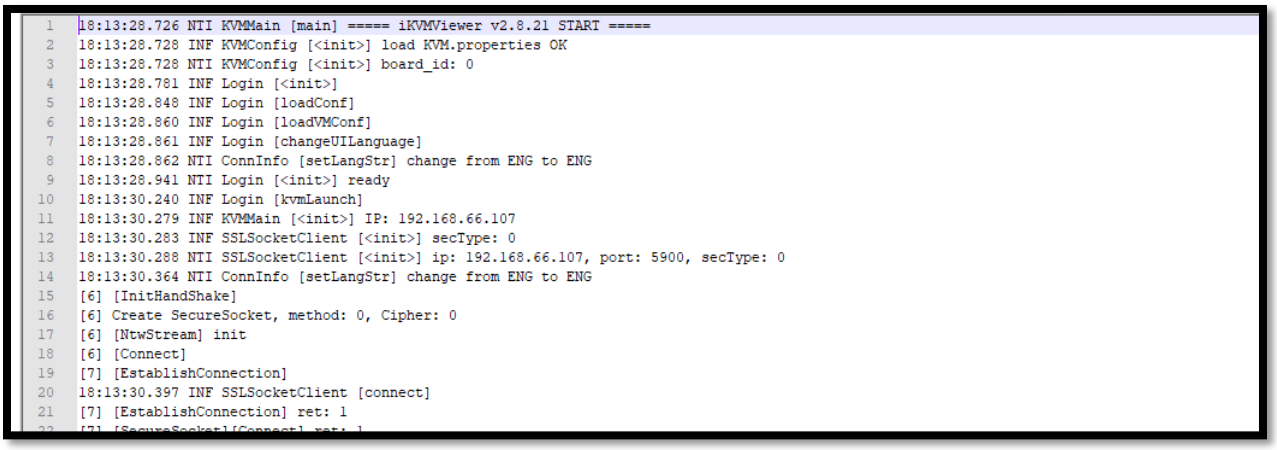
When users want to debug the Java Client iKVMViewer.jar, they can change the log level to obtain more detailed debug information. Click the "Debug Log" tab to enter the settings page.

- Setting **"Log Level"**: DISABLED, EMERGENCY, ALERT, CRITICAL, ERROR, WARNING, NOTICE, INFO or DEBUG.



- Use console to launch iKVMViewer.jar to get the log message.

```
java -jar iKVMViewer.jar
```



```
18:13:28.726 NTI KVMMain [main] ===== iKVMViewer v2.8.21 START =====
18:13:28.728 INF KVMConfig [<init>] load KVM.properties OK
18:13:28.728 NTI KVMConfig [<init>] board_id: 0
18:13:28.781 INF Login [<init>]
18:13:28.848 INF Login [loadConf]
18:13:28.860 INF Login [loadVMConf]
18:13:28.861 INF Login [changeUILanguage]
18:13:28.862 NTI ConnInfo [setLangStr] change from ENG to ENG
18:13:28.941 NTI Login [<init>] ready
18:13:30.240 INF Login [kvmLaunch]
18:13:30.279 INF KVMMain [<init>] IP: 192.168.66.107
18:13:30.283 INF SSLSocketClient [<init>] secType: 0
18:13:30.288 NTI SSLSocketClient [<init>] ip: 192.168.66.107, port: 5900, secType: 0
18:13:30.364 NTI ConnInfo [setLangStr] change from ENG to ENG
[6] [InitHandShake]
[6] Create SecureSocket, method: 0, Cipher: 0
[6] [NtwStream] init
[6] [Connect]
[7] [EstablishConnection]
18:13:30.397 INF SSLSocketClient [connect]
[7] [EstablishConnection] ret: 1
```

3.3.3 Full-Screen Mode

Click the "Full-Screen Mode" button to enable/disable Full-Screen.

3.3.4 OSD UI Style

This feature will change the menu bar to the following style.

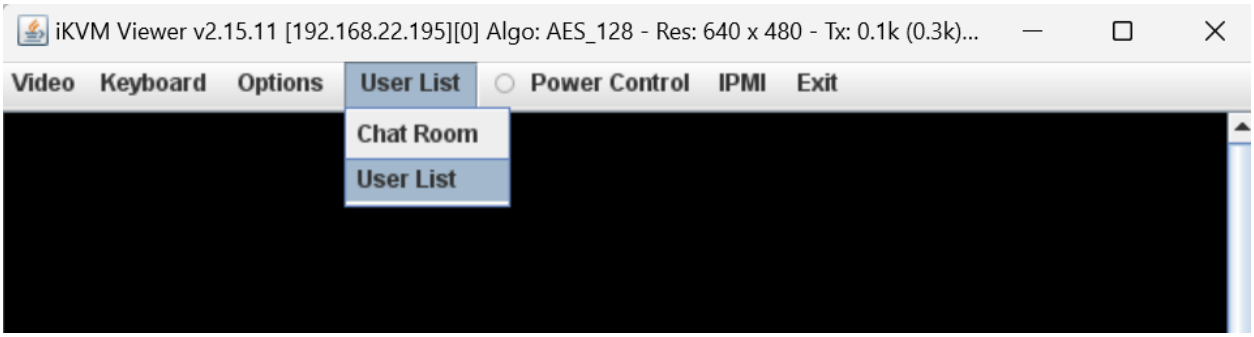
- It includes some Common Functions.

- : Move OSD
- : Hotkey Settings
- : Virtual Storage
- : Virtual Keyboard
- : Preference
- : Full-Screen Mode
- : Exit
- : Show User List
- : Menu bar UI Style
- : Keyboard Mouse Hotplug
- : Key-Macro
- : Record
- : Power Control

3.3.5 Keyboard Mouse Hotplug

- Click to activate Keyboard Mouse Hotplug. Sometimes the USB device will be disconnected from the Host. Use hotplug to re-register the BMC as keyboard/mouse USB to Host.

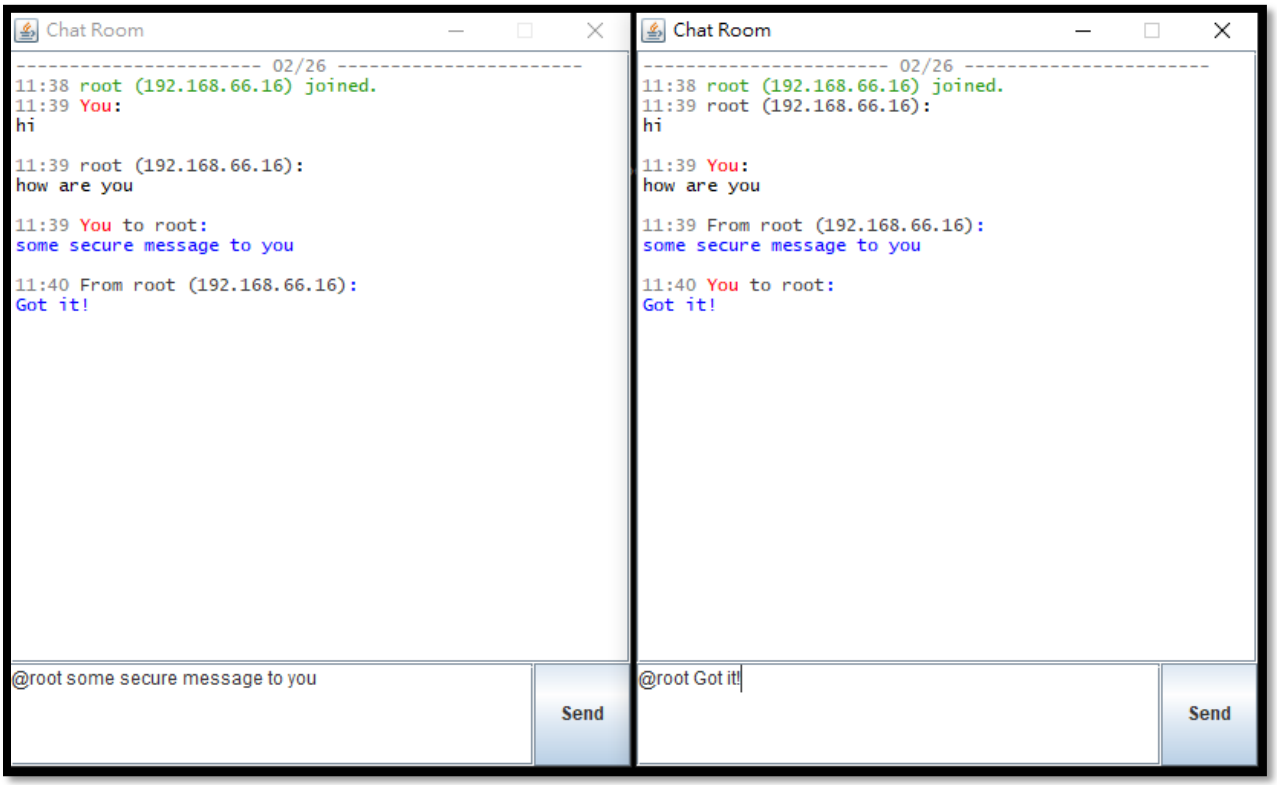
3.4 User List



3.4.1 Chat Room

Provide a simple instant messaging communication method between clients.

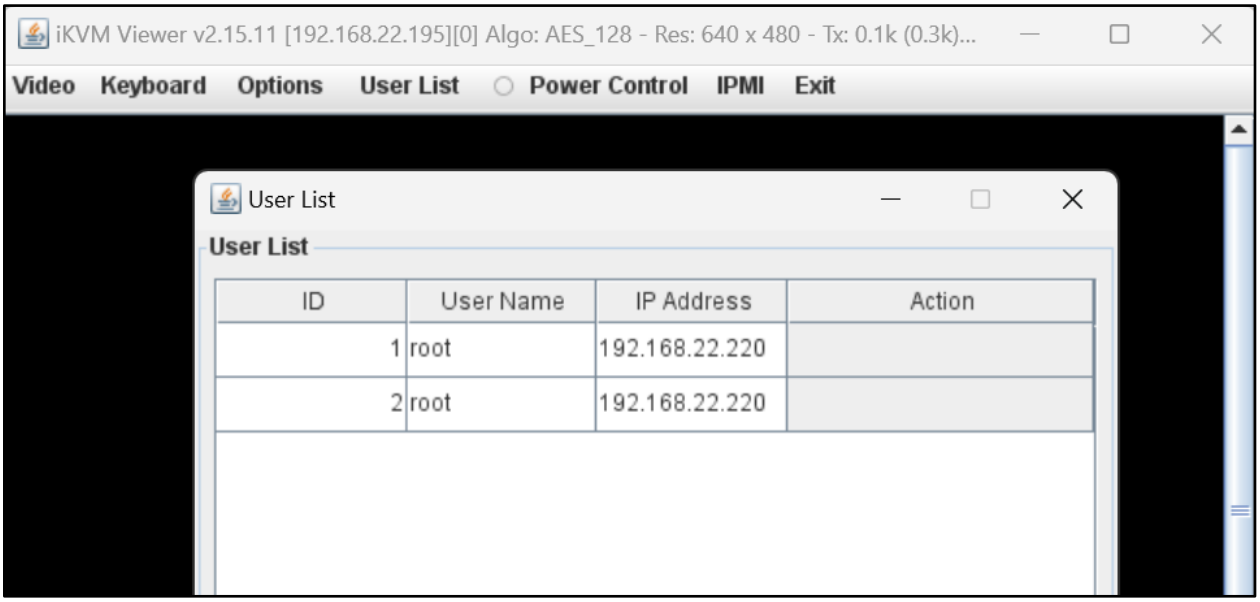
- You can see who joined/left the chatroom.
- When you say something, your messages are visible to everyone if you don't use the whisper format.
- If you want to whisper to someone. Use the format: "@USERNAME*WHAT-YOU-SAY*".
- **Notice**: the maximum length of input text area is 500 characters.



3.4.2 User List

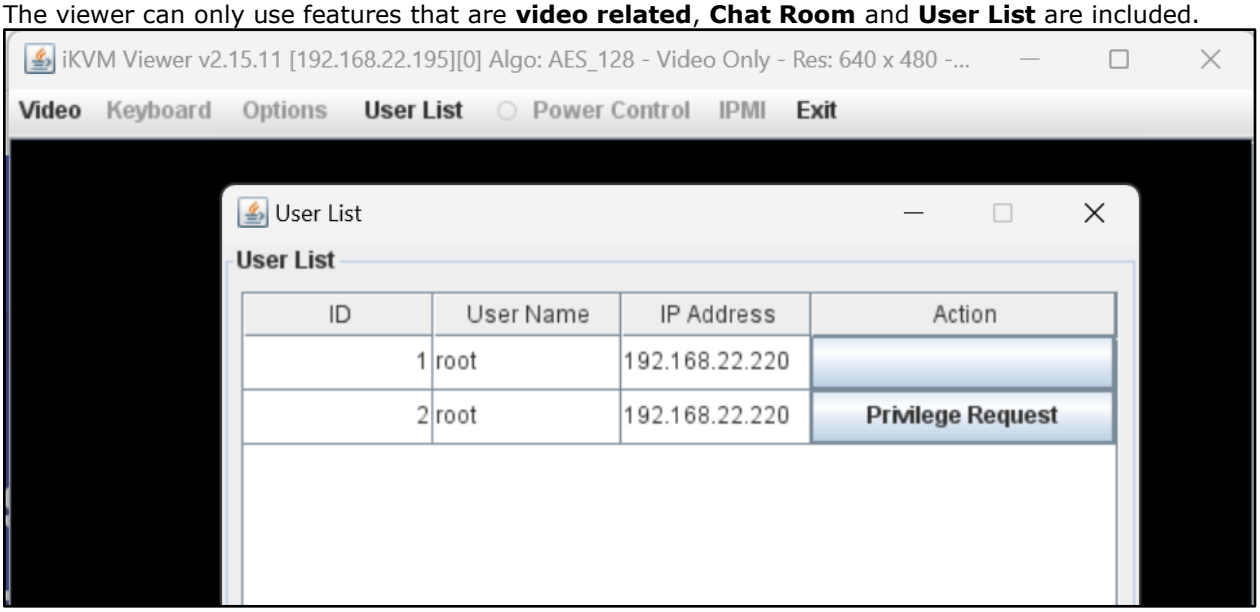
User list provides a list to let users know who uses KVM service. The first one (named owner) will get the keyboard/mouse privilege. The others (named viewer) can only view the Host screen.

- The owner



- The viewer

The viewer can only use features that are **video related**, **Chat Room** and **User List** are included.



- When the viewer wants to receive privilege, he can follow the privilege request flow:

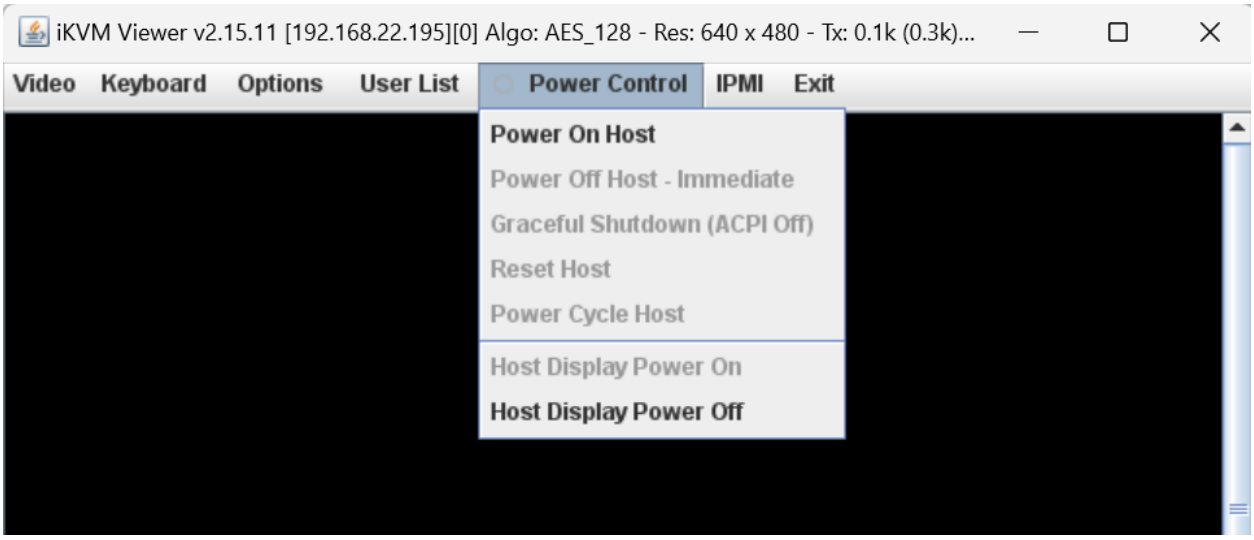
1. The viewer presses the **privilege request button** to send the request to the owner.
2. When the owner receives the request, a message bar display to notify owner for action.

3.1 Press **Reject**. Nothing happens. But the requester will get a reject response on the message bar.

3.2 Press **Accept**. The requester will get a notification and become a new owner, and the previous owner will become a viewer.

3.5 Power Control

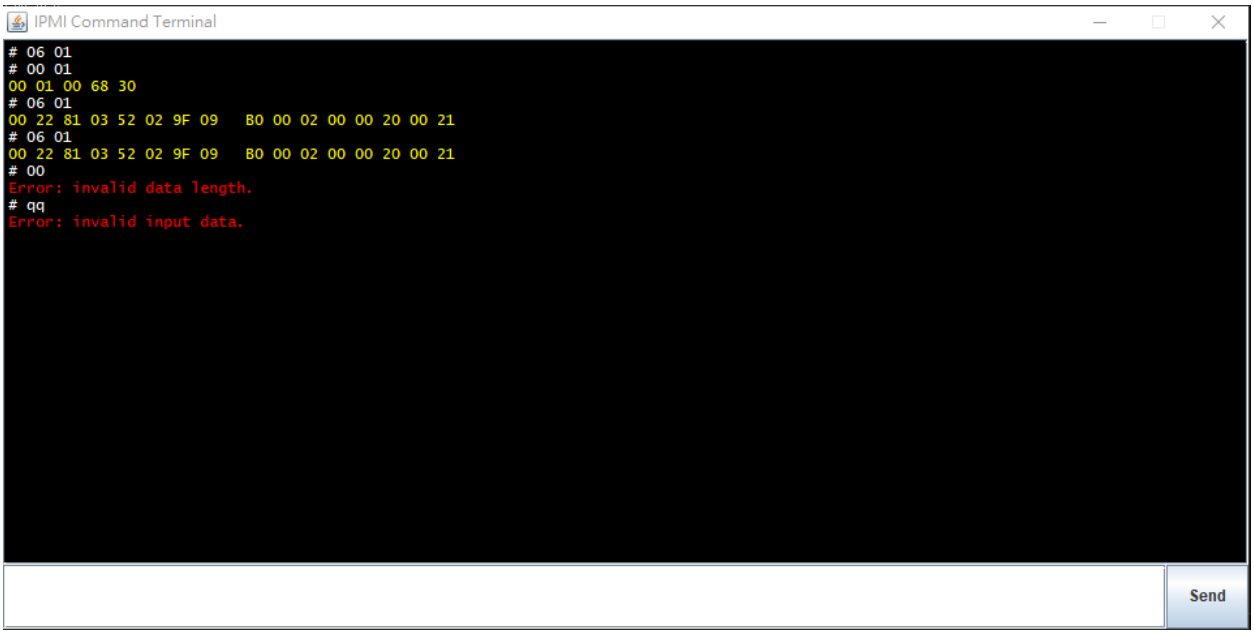
Control the remote host's power and display.



- Power control:
  - **Power ON Host** - Power On the Host CRB Board.
  - **Power OFF Host** - Power Off the Host CRB Board (Sometimes will be blocked by OS).
  - **Gracefully Shutdown (ACPI Off)** - Emulates OS Software Reset. Let OS show an interaction UI to void shutdown suddenly.
  - **Reset Host** - Reset the Host CRB Board (Sometimes will be blocked by OS).
  - **(soft) Power Cycle** - Power Cycle Host CRB Board. The shutdown action is like ACPI Off and then power on automatically.
- Display power control:
  - **Host Display Power ON** - Power on host display.
  - **Host Display Power Off** - Power off host display.
- Note that:
  - On ikvmserver, the power cycle feature uses the **0x00 0x02 0x0E ipmi command** rather than **0x00 0x02 0x02 ipmi command**. The latter will harm the disk when run on OS environment.

3.6 IPMI Command Window

- Users can input ipmi command in hex format. And then get the response in hex format.
- When a command is sent, the "Send" button will be blocked until a response is received.



- **Notice**: the maximum length of input text area is 500 characters.

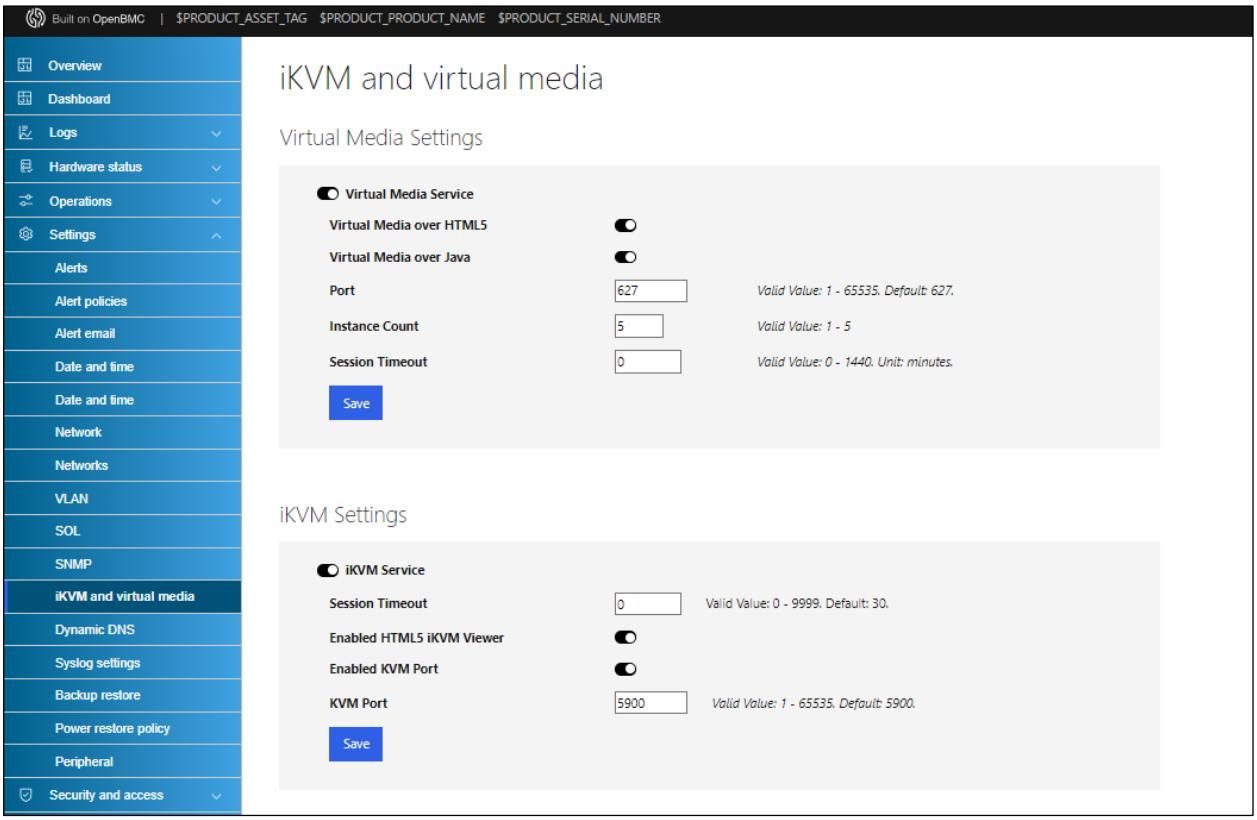
3.7 Exit

- Exit the Java KVM viewer application.

4 Known Issues

4.1 Failed to connect ikvmserver

- Make sure the server is alive.
  - Browse the BMC web page: WebUI > Remote Control > iKVM
  - Checkout the Launch button is available.
- Make sure the port (Ex: 5900) is matched.



4.2 Key event blocked by input method

- The key event will be blocked by the input method on the OS that makes abnormal input behavior. It is suggested to use the default ENG input method to operate IKVMViewer.
- Known input method such as:
  - Windows 注音輸入法
  - ubuntu 注音輸入法

4.3 Install Japanese Input Method

Install language package on the Operating system.

- Windows 10
    a. Win menu > Setting > Region & Language > Language > Add a preferred language
    b. Select 日本語 and install it
    c. Select 日本語 and click options
    d. Click "change keyboard layout" to Japanese keyboard 106/109
- Ubuntu 18.04
    a. Setting > Language support > Install Language > Japanese
    b. Setting > Region & Language > Input Source > Add Japanese
    c. open terminal to adjust keyboard layout
        - sudo dpkg-reconfigure keyboard-configuration
        - Sun Type 6 (Japanese) > Japanese > Japanese (Kana)
    - Install input method (hiragano / katagana)
        - Hiragano: 平假名
        - katagana: 片假名

4.4 The mouse cursor disappears in Ubuntu 16.04

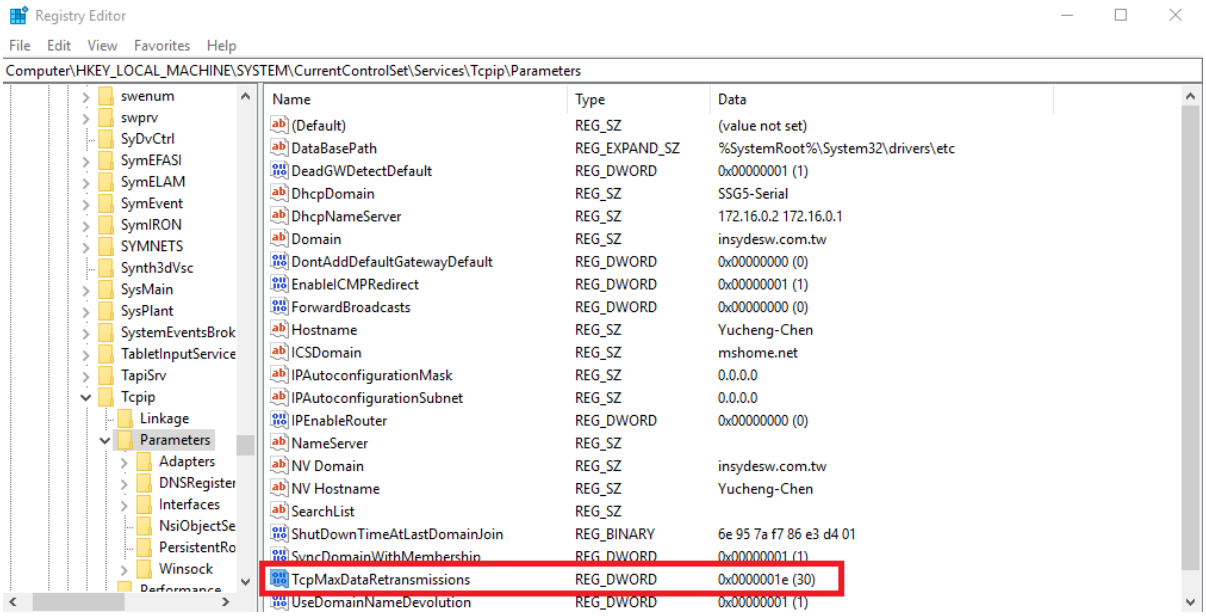
- Due to GNOME's cursor plug-in.
- To fix this issue, please disable the cursor plug-in by executing the following command on Ubuntu 16.04:

```
gsettings set org.gnome.settings-daemon.plugins.cursor active false
```

4.5 Connection Timeout

- Normally we have 30s TCP socket timeout, which means you could continue the connection if you lost the connection less than 30s ago.
- However, due to the Windows OS environment limitation, the TCP timeout time is limited (about 20s). Modify it to reach 30 seconds timeout.
    - Method: (Reference: https://deploymentresearch.com/research/post/532/fix-for-windows-10-exhausted-pool-of-tcp-ip-ports)
        - Open run command (Win +R)

- Key in "regedit" and press enter (with Administrator privilege)
- Find "Computer\HKEY_LOCAL_MACHINE\SYSTEM\CurrentControlSet\Services\Tcpip\Parameters"
- Add a new DWORD value "TcpMaxDataRetransmissions (REG_DWORD) = 30" (decimal)
- Reboot



- If you unplug the network wire to test it, you should see the following notices.
  - OS will detect that the wire is unplugged, and will force to reset the TCP socket. You should not unplug the one between the OS and the switch.
  - When you plug the network wire back in, the connection won't be established immediately. It may take several seconds to recover, and that counts towards the timeout timer (in seconds).

4.6 Unable double clicks to run .jar

- on Linux like platform: "It may be dangerous to run. Read about the executable bit"
  - This is a security issue in Linux. After downloading a .jar, the user should manually enable the execute bit. And then the .jar file can be pressed and run.
  - or... run on command line (terminal).

```
java -jar iKVMViewer.jar
```

4.7 About Video Only

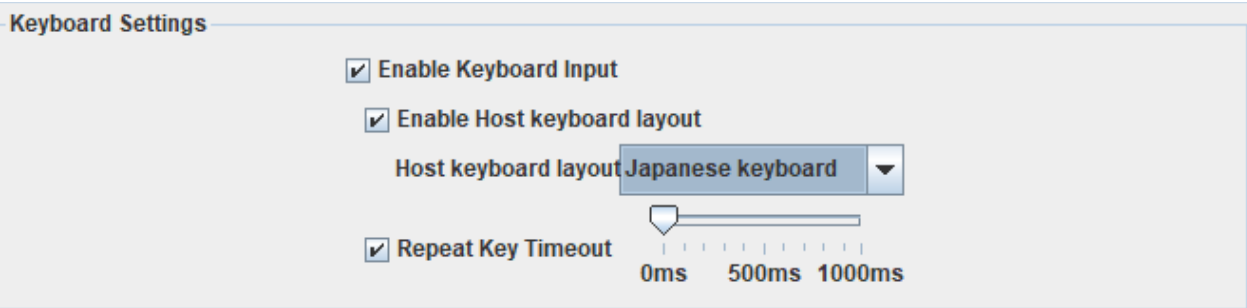
- The first one connected is the named owner who is able to do all operations.

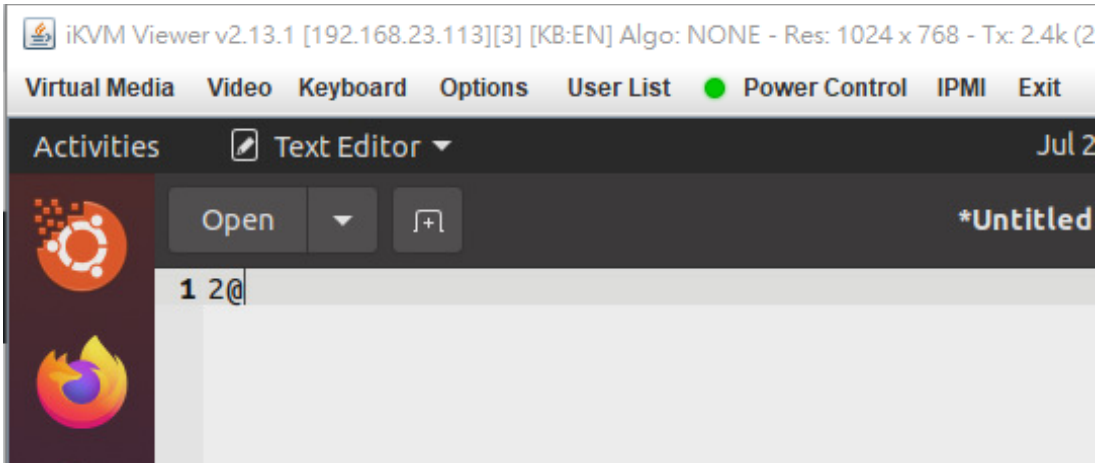
- The other connected is named as video-only who can view video only.
- The available features for different privileges.

Item	Master	Slave (Video Only)
Get Video Frame	V	V
Send Keyboard Event	V	
Send Mouse Event	V	
Power Control	V	
IPMI Command	V	
Advance Setting	V	
Virtual Storage	V	

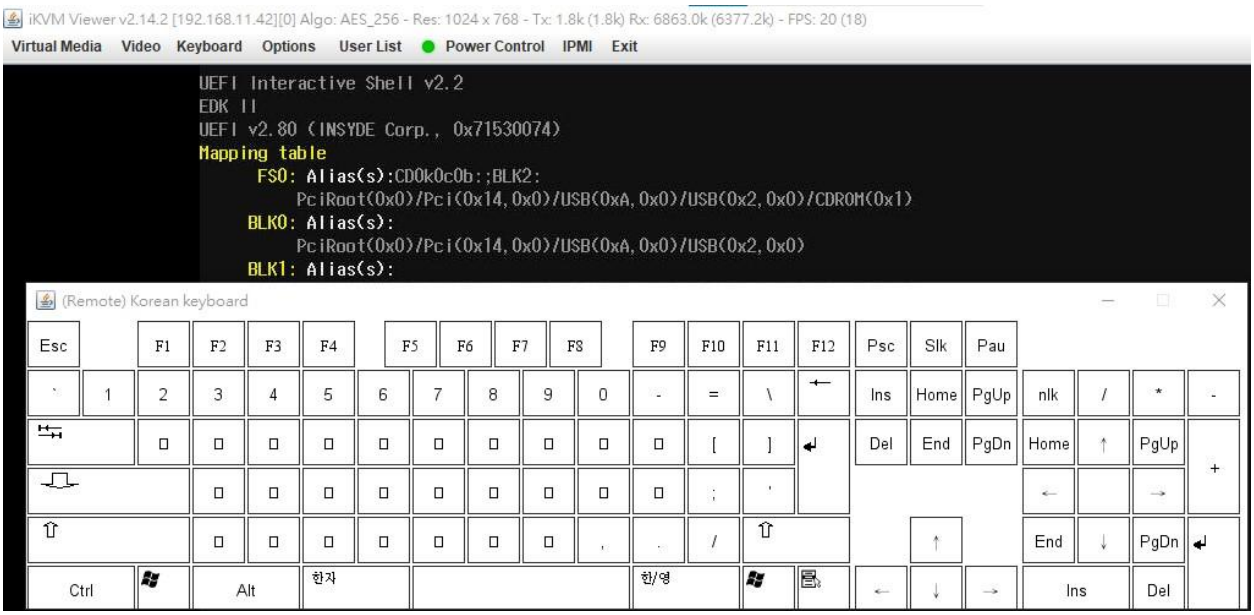
4.8 Host Keyboard Layout Mapping Feature

- Usage:
  - Mapping the keyboard layout between Host and client keyboard. Let users key with his own keyboard layout character and output the same character on Host machine.
- Prepare:
  - Setup the **Host Keyboard Layout** as what keyboard layout the Host OS usage.
- Example: If the ***host-side keyboard layout is*** *JP* ***and the*** *client-side* keyboard layout is EN.
  - Java Viewer > Menu > Optional > Preference > Input. Setup the **Host Keyboard Layout** as "Japanese"
  - Host side open text editor such as **gedit** or **notpad**.
  - On the client-side, user type "2" and the Host side will show character "2".
  - On the client-side, user type "@" (shift+2) and the Host side will show character "@"。



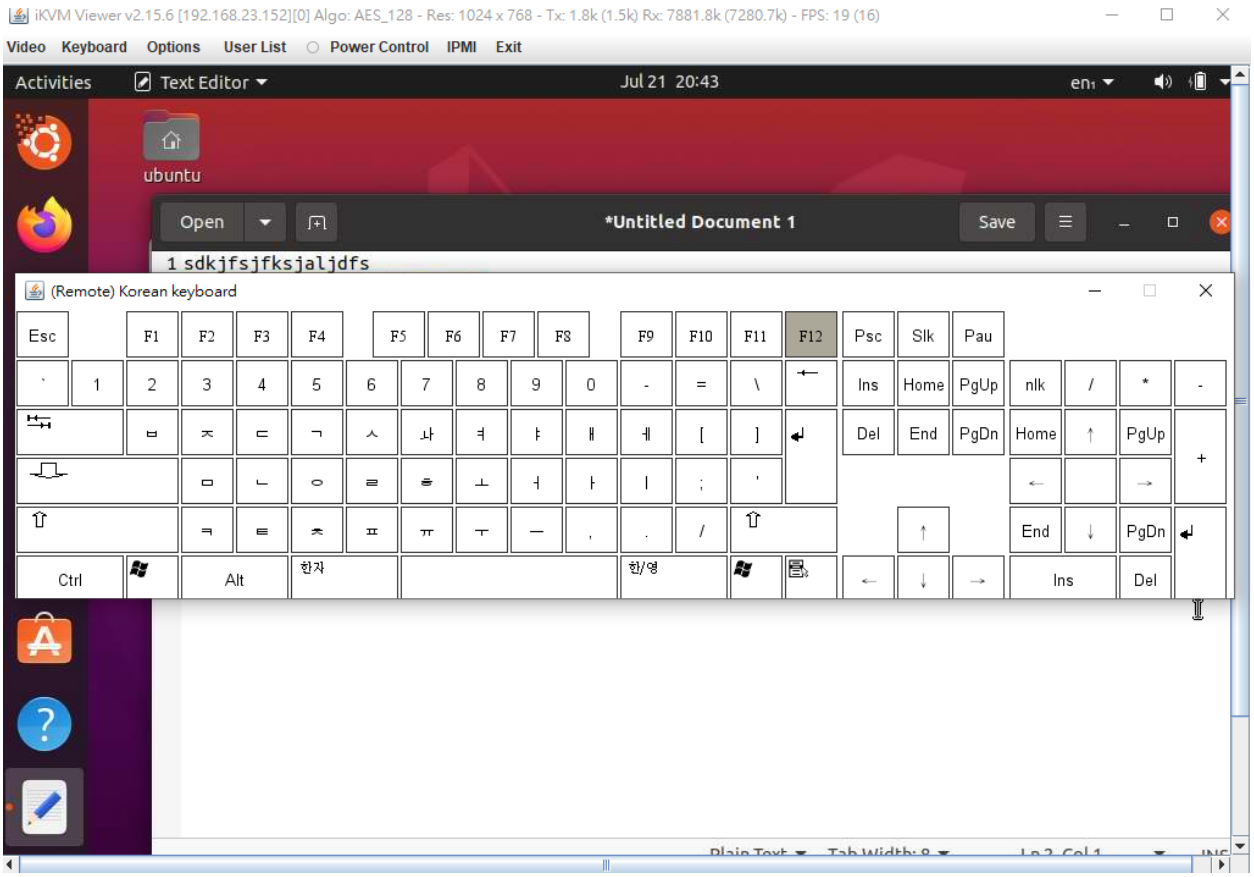


4.9 Virtual Keyboard Unable to Show Korea Font



korea (client OS) 需要安裝 korea 字形, 按鍵的圖樣才會正常顯示

設定
首頁
尋找設定
時間與語言
日期和時間
地區
語言
語音
語言
中文(台灣)
鍵盤
繁體中文 (台灣)
Windows 顯示
中文(台灣)
請如 [設定] 和 [檔案
慣用語言
應用程式及網站將
新增語言
繁體中文 (台
選擇要安裝的語言
韓文
한국어
韓文
顯示語言
文字轉換語音
語音辨識
手寫
下一步
取消


iKVM Viewer v2.15.6 [192.168.23.152][0] Algo: AES_128 - Res: 1024 x 768 - Tx: 1.8k (1.5k) Rx: 7881.8k (7280.7k) - FPS: 19 (16)
Video Keyboard Options User List Power Control IPMI Exit
Activities Text Editor Jul 21 20:43
ubuntu
Open *Untitled Document 1 Save
1 sdkjfsjfksjaljdfs
(Remote) Korean keyboard